

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS

COURSE CODE: CSA0706

COURSE OUTCOME COVERED

C04: Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards

(BL2, BL6)

Assessment Tool Weightage: 30%

QUESTIONS: 5

Total Marks:

25

Annexure A

Comparative Analysis

Code of Conduct:

*I _____V.Lasya(192524186)_____ (Name / Reg No)
certify that this submission is my original work and that I have adhered
to the guidelines specified for this assessment.*

*I understand that any violation of academic integrity rules will result in
disciplinary action.*

Signature of the student: v.lasya

Annexure A

Comparative Analysis

1.

Star topology is generally more expensive than bus topology because it requires a central device such as a switch or hub and more cabling to connect each node individually. In contrast, bus topology uses a single backbone cable, making it low-cost and easier to install. However, in terms of reliability, star topology is superior because failure of one node does not affect others, whereas a failure in the main bus cable can bring down the entire network. Performance is also better in star topology since data transmission is managed through a central device, reducing collisions, while bus topology suffers from performance degradation as more devices are added due to increased data traffic on the shared medium.

ANSWER :

Cost

Star topology requires a dedicated switch or hub as the central connecting device, and every single node needs its own separate cable run back to that device. In a large office with many workstations, this can mean hundreds of metres of extra cabling and the added expense of networking

hardware. Bus topology, by contrast, uses only one continuous backbone cable to which all devices are tapped, so it needs far less cable and no central hardware, making it the cheaper option for small, budget-constrained networks.

Reliability

Since each device in a star network has its own dedicated link to the central switch, the failure of one cable or one node has no effect on the rest of the network — only that single device loses connectivity. Bus topology, however, depends entirely on the health of the single shared backbone; if the main cable breaks, gets damaged, or a connector fails, the entire network can go down because every device shares that one physical medium.

Performance

A central switch in star topology manages and forwards traffic intelligently to only the intended recipient, so collisions are minimised and the effective bandwidth available to each device stays high even as the network grows. Bus topology suffers here because all devices share the same cable and the same bandwidth; as more devices are added, the chance of data collisions increases and overall network performance degrades noticeably.

Conclusion

Bus topology wins on low initial cost and simplicity, while star topology is the stronger choice wherever reliability and consistent performance matter more than upfront expense — which is why star topology is the dominant choice in almost all modern LAN installations.

2.

Mesh topology offers very high fault tolerance because every node is connected to multiple other nodes, providing multiple paths for data transmission. If one link fails, data can still be routed through alternate paths without disrupting communication. In contrast, star topology has moderate fault tolerance; while failure of individual nodes does not affect the network, failure of the central hub or switch can cause the entire network to stop functioning. Therefore, mesh topology is more robust and reliable in critical environments compared to star topology.

ANSWER :

Fault Tolerance in Mesh Topology

In a full mesh, every node is connected to every other node, and in a partial mesh most nodes have more than one connection. This redundancy means that if any single link or even several links fail, data can automatically be rerouted through one of the many alternate paths that still exist between source and destination, which is why mesh designs are chosen for mission-critical infrastructure such as ISP backbones, military

communication systems, and industrial control systems where downtime is unacceptable.

Fault Tolerance in Star Topology

Star topology relies on a single central device through which all communication must pass. If an individual node or its cable fails, only that one device is affected — similar to mesh. However, the central device itself is a single point of failure: if the switch or hub stops working, every connected device loses network access simultaneously, bringing the entire network down at once.

Trade-offs

The cost and complexity are the trade-off for mesh topology's superior fault tolerance. The number of cable connections required in a full mesh grows very quickly (following $n(n-1)/2$ for n nodes), making it expensive and difficult to scale. Star topology avoids this problem, and its single point of failure risk can be reduced in practice by using a redundant or backup switch.

Conclusion

Mesh topology is clearly superior in fault tolerance and is the preferred choice for critical environments that cannot tolerate downtime, whereas star topology offers a reasonable, more economical balance of manageability and resilience for everyday business and home networks.

3.

Bus topology is simple and easy to install because it uses a single backbone cable with minimal connections, making it suitable for small networks. On the other hand, mesh topology is highly complex to install as each node must be connected to every other node, requiring a large number of cables and ports. This complexity increases significantly as the number of devices grows, making mesh topology time-consuming and costly to implement compared to the straightforward setup of bus topology.

ANSWER :

Installation of Bus Topology

Bus topology requires only a single backbone cable running through the area to be networked, with each device connected to it through a simple tap or connector. There is no need for a central switching device, and the amount of cabling required is minimal since devices share one common line, making it quick to deploy for small, temporary, or low-budget networks.

Installation of Mesh Topology

Because every node must be directly connected to every other node (in a full mesh) or several other nodes (in a partial mesh), the number of physical cable runs and ports required increases dramatically as more

devices are added. Connecting just 10 devices in a full mesh requires 45 separate cable links, compared to a single backbone cable in bus topology, meaning mesh installations demand significantly more planning, hardware, labour, and time.

Real-World Impact

Bus topology can be set up by a single technician in a short time with basic tools, while a mesh network of any significant size typically requires a dedicated project team, structured cabling plans, and considerably more budget. This is why mesh topology is generally reserved for smaller, high-value critical segments of a network (such as core routers) rather than connecting large numbers of end-user devices directly.

Conclusion

Bus topology is clearly the simpler and faster topology to install, while mesh topology, despite its excellent fault tolerance, comes at the cost of substantially higher installation complexity, time, and expense.

4.

Hybrid topology is highly scalable because it combines two or more different topologies (such as star, mesh, or bus), allowing the network to expand flexibly according to organizational needs. New segments can be added without affecting the existing network structure. In comparison, star topology is also scalable but to a limited extent, as it depends on the

capacity of the central device (switch or hub). Once the central device reaches its port limit, expansion requires additional hardware. Thus, hybrid topology provides greater scalability than star topology.

ANSWER :

Scalability of Hybrid Topology

Because hybrid topology combines multiple basic topologies into one larger network, new segments, departments, or entire sub-networks can be added as the organisation grows without tearing down or redesigning the parts that already exist. For example, a company could connect several star-topology department LANs through a mesh-topology core backbone, then simply plug in a new department whenever it opens, with minimal disruption to everyone else — which is why hybrid networks are the standard choice for medium-to-large organisations and campus networks.

Scalability of Star Topology

Adding a new device to a star network is simple — it just needs a cable run to the central switch — but this is directly limited by the port capacity of that device. Once all ports on the switch or hub are used, further expansion requires purchasing and installing additional switches, and potentially restructuring the cabling layout, which adds cost and complexity.

Structural Flexibility

Hybrid topology can adapt its very structure to fit changing needs (mixing reliability-focused mesh segments with cost-effective star segments, for instance), while star topology's structure stays fixed around one central device, so its growth is constrained to expanding within that same basic design.

Conclusion

Hybrid topology offers far greater and more flexible scalability because it is not tied to a single design and can incorporate new topologies and segments as needed, whereas star topology can grow only in a linear, hardware-limited way around its central switch.

5.

In terms of maintenance, star topology is relatively easy to manage because issues can be quickly identified and isolated to a specific node or cable. Mesh topology, while highly reliable, is difficult to maintain due to its complex structure and large number of connections, making troubleshooting time-consuming. Bus topology is harder to maintain than star because faults in the backbone cable can disrupt the entire network and are difficult to locate. Hybrid topology has moderate maintenance complexity; while it benefits from flexibility, managing multiple combined topologies requires careful monitoring and skilled administration.

ANSWER :

Maintenance of Star Topology

Because every device has its own dedicated cable back to the central switch, a fault is usually confined to a single, easily identifiable link, so administrators can quickly pinpoint which cable or node is causing an issue and fix it without disturbing the rest of the network. Centralised monitoring through the switch also makes it easy to see the status of every connected device from one place.

Maintenance of Bus Topology

Since all devices share one backbone cable, a fault anywhere along that cable can disrupt the entire network, and because the cable often runs through walls, ceilings, or long distances, physically locating the exact point of failure can be time-consuming and may require specialised cable-testing equipment.

Maintenance of Mesh Topology

Despite being highly reliable due to its redundant links, mesh topology is the most difficult to maintain day-to-day. The sheer number of connections and cables involved means that tracking, labelling, and troubleshooting requires detailed documentation and considerable administrative effort, since identifying which of many overlapping paths is causing a performance issue is inherently complex.

Maintenance of Hybrid Topology

Because hybrid topology is composed of multiple combined topologies, maintenance complexity depends on which part of the network is affected — a fault in a star segment is as easy to resolve as pure star topology, while a fault in a mesh segment carries the same difficulty as pure mesh. Hybrid networks therefore require skilled administrators who understand each section and can apply the correct troubleshooting approach, along with careful, ongoing monitoring across the whole combined structure.

Conclusion

Star topology offers the easiest maintenance, bus topology is moderately difficult due to shared-cable fault-finding, mesh topology is the most maintenance-intensive because of its complexity, and hybrid topology requires a balanced, moderate level of skilled and careful administration across its combined segments.

RUBRICS FOR CALCULATION

Criteria	Weightage	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Understanding of Concepts	15%	Demonstrates complete and in-depth understanding of all concepts being compared	Good understanding with minor gaps	Basic understanding; some concepts unclear	Poor understanding of concepts
Comparison Depth	20%	Provides detailed, insightful, and meaningful comparisons across multiple aspects	Adequate comparison with relevant points	Limited comparison; lacks depth	Minimal or incorrect comparison
Criteria Selection	10%	Selects highly relevant and appropriate comparison parameters	Mostly relevant criteria	Some irrelevant or missing criteria	Poor or no criteria selection
Analytical Skills	15%	Strong analysis with logical reasoning and clear justification	Good analysis with some justification	Basic analysis with weak reasoning	No clear analysis
Organization & Structure	10%	Well-organized, clear, and logically structured	Generally organized	Somewhat disorganized	Poorly structured
Use of Examples / Evidence	10%	Uses strong, relevant examples or data to support comparison	Uses some examples	Limited examples	No supporting examples
Clarity of Presentation	10%	Very clear, concise, and easy to understand	Mostly clear	Somewhat unclear	Difficult to understand

Conclusion & Insights	10%	Insightful conclusion with strong justification	Clear conclusion	Basic conclusion	No or weak conclusion
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